CSAI 252 Introduction to Computer Networks

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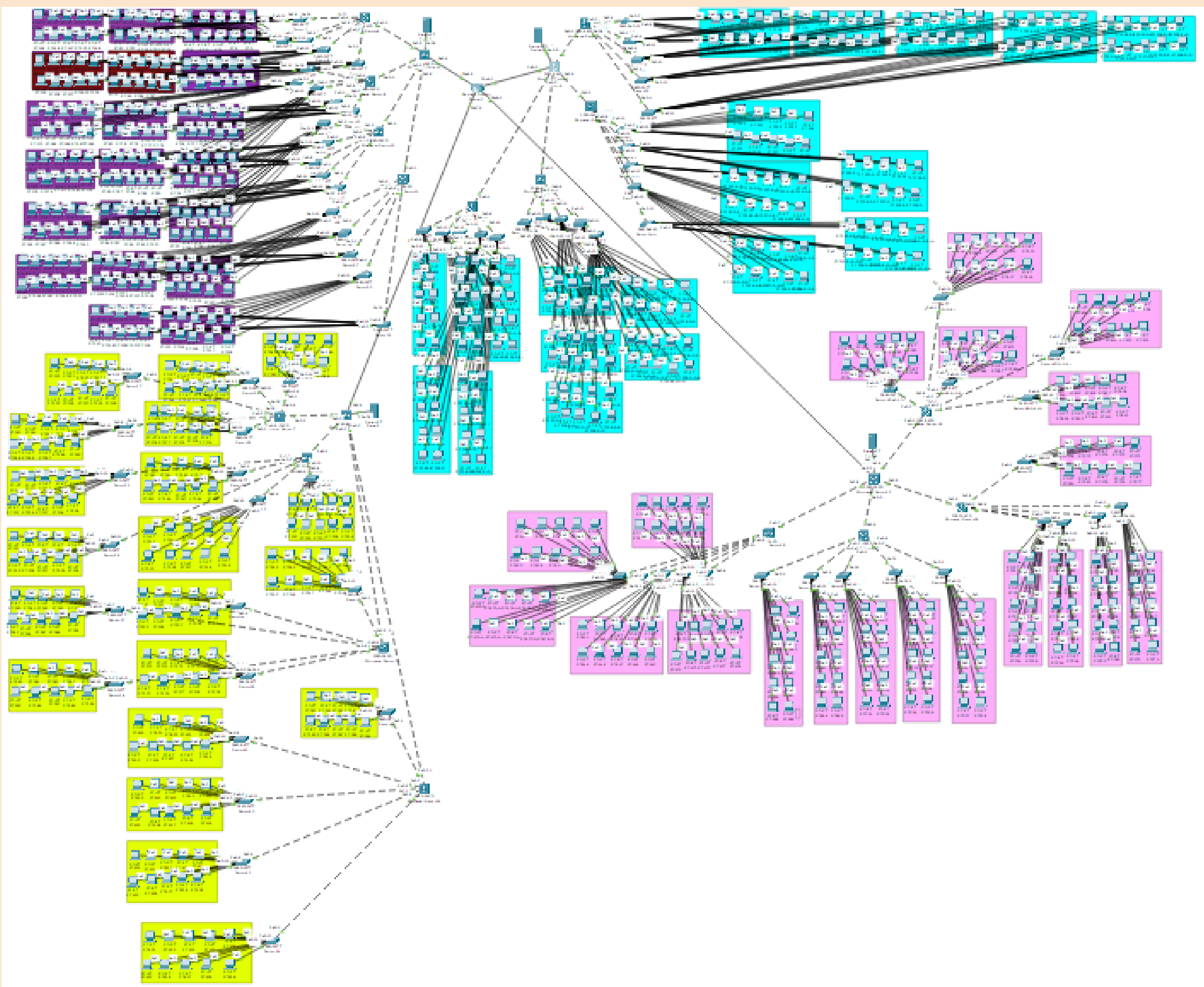
Overview:

Our project is a university network that connects the Computers in labs together using switches, router, and servers. The network topology was implemented on Cisco packet tracer to visualize how each device is connected to the others. The network connects four faculties and ensures the connectivity between devices and sub-networks. In addition, we implemented classless addressing to distribute by IP using Dynamic Host Configuration Protocol.

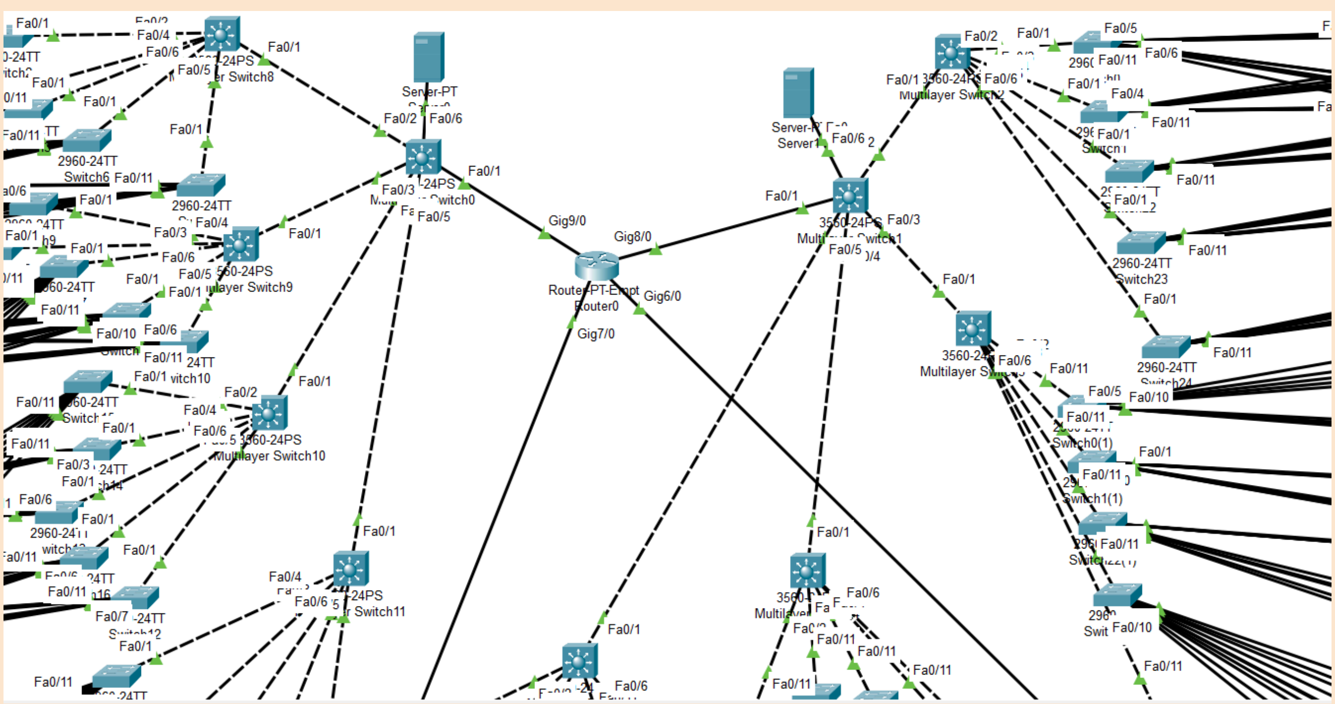
TOPOLOGY

We used Cisco Packet Tracer to implement the entire university network. We are four members in the team so, each one of us worked on creating one faculty devices. In each faculty(multiswitch), there are four floors(multiswitches). On each floor(multiswitch), there are five labs(switches). Each lab(switch) includes ten PCs. To clearly specify the number of PCs in each lab. We grouped each ten PCs in a colored rectangle.



Router Connections

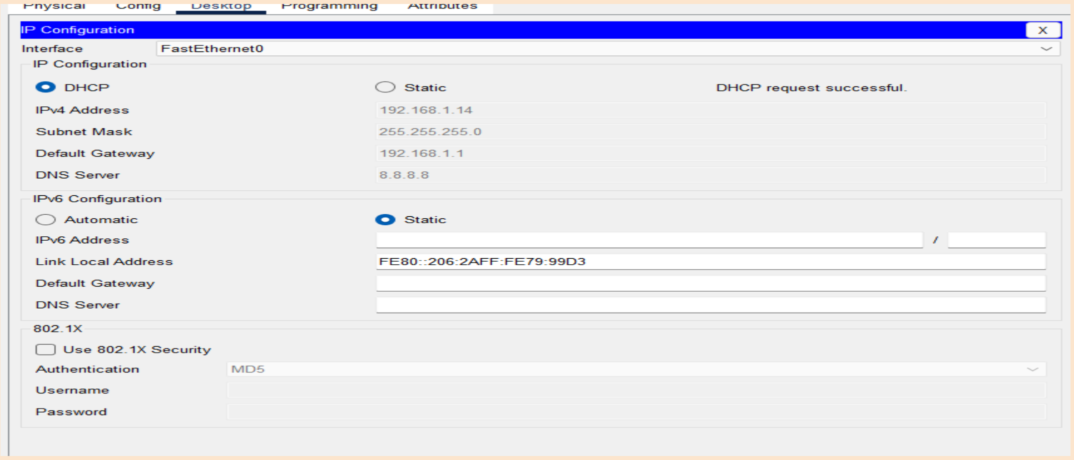
The university has one router that connects the four faculties (multiswitches) together. We applied a star topology connection as we believe this is the best topology type to implement in a university campus with eight hundred PCs. Star topology has many advantages. First, it is safe. The failure of one device connection will not lead to the damage of the entire network because each device has its own connection. Second, it is secure. The devices are connected individually therefore, it is difficult for unauthorized users to access the network. Third, it is flexible. It can accommodate a variety of cable types according to the network needs and budget.



DHCP

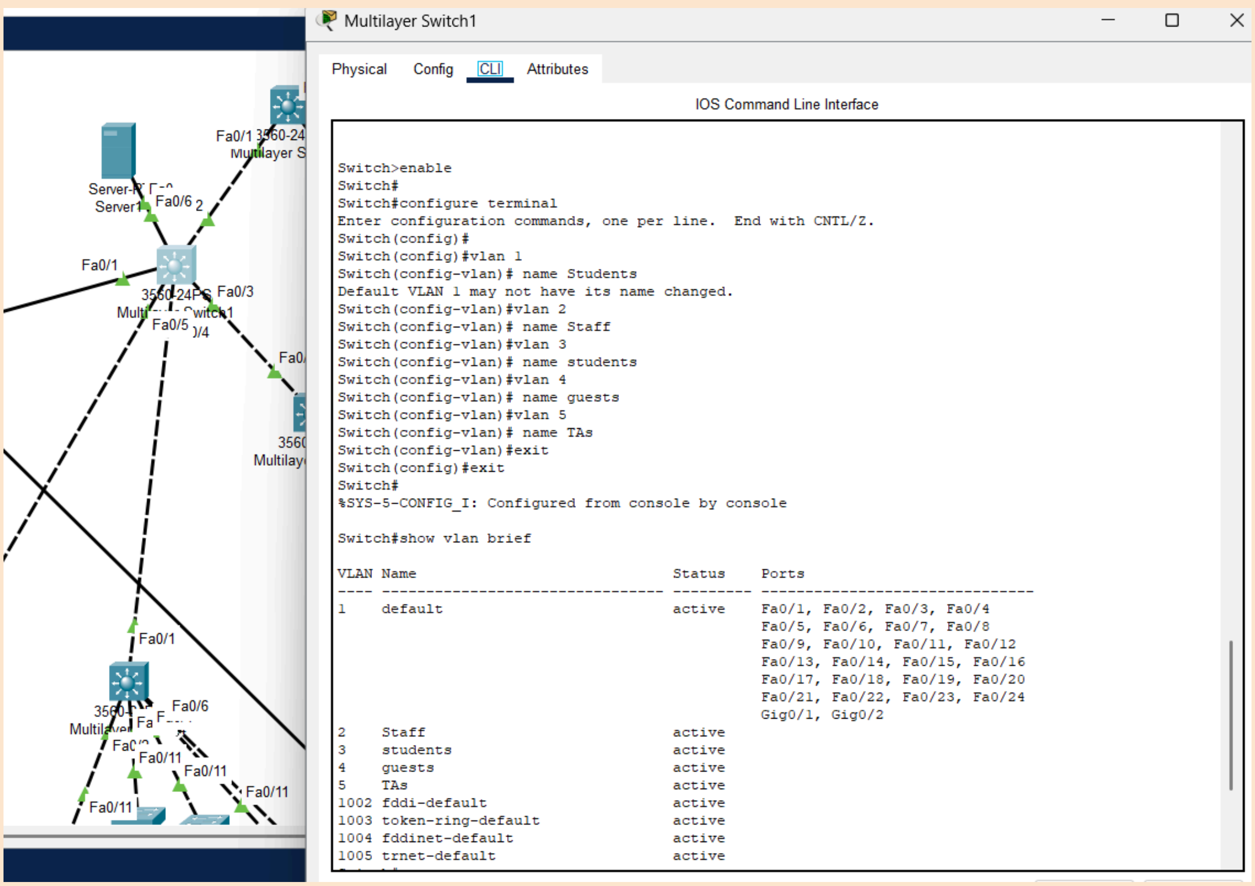
To implement the Dynamic Host Configuration Protocol (DHCP), we started by writing the network, ip address, default router, and dns server for each faculty in the router command. This resulted in creating an IP for each PC by DHCP. Then, We changed each PC from Static IP Configuration to dynamic IP configuration. Finally, the IP addresses were created dynamically for each PC.

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp excluded-address 192.168.3.1 192.168.3.10
ip dhcp excluded-address 192.168.4.1 192.168.4.10
!
ip dhcp pool LAN1
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
ip dhcp pool LAN2
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 8.8.8.8
ip dhcp pool LAN3
 network 192.168.3.0 255.255.255.0
 default-router 192.168.3.1
 dns-server 8.8.8.8
ip dhcp pool LAN4
 network 192.168.4.0 255.255.255.0
 default-router 192.168.4.1
 dns-server 8.8.8.8
```



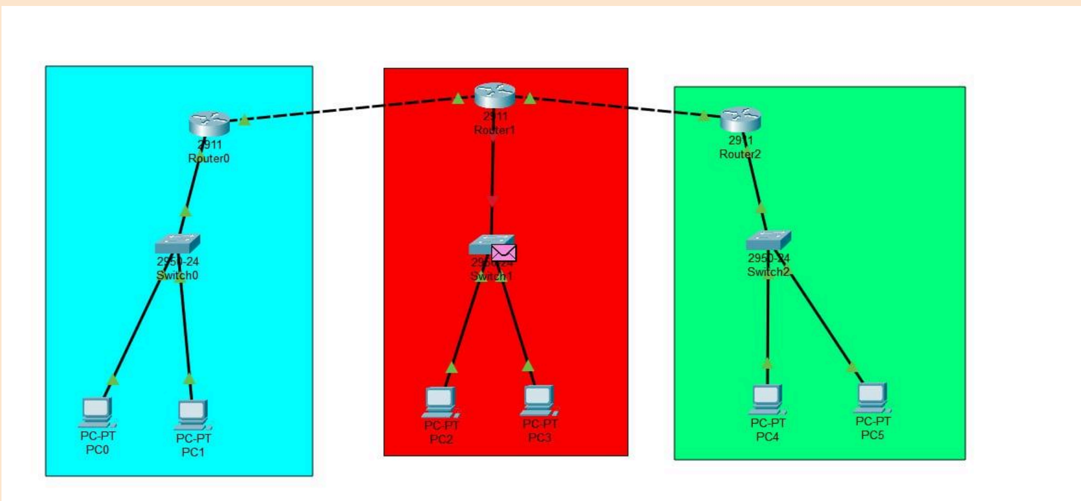
VLAN

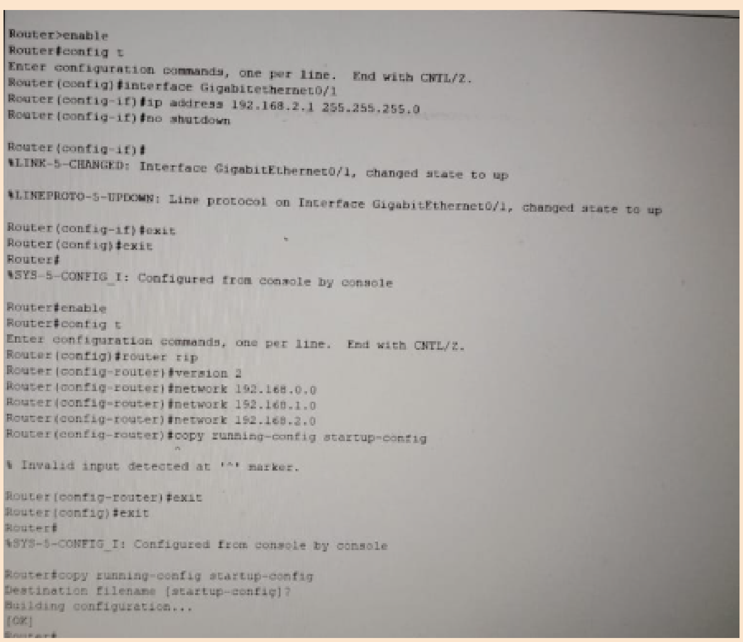
The Virtual Local Area Network is applied on a faculty (multiswitch). Generally, VLAN is used to connect multiple devices and network nodes from different LANs into one logical network. Moreover, it improves security and traffic management in the network.



RIP

Routing Information Protocol (RIP) is generally used to manage the routing in the university network. It is implemented by configuring the main router and switches to use RIP for routing data between faculties and the Scientific Computation Center.





```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Gigabitethernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router(config-router)#copy running-config startup-config
                      ^
% Invalid input detected at '^' marker.

Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```